

Dimension Reduction Analysis: pathmnist

Generated report #2

Seed: 0 | Plan revision: 1

Dataset profile & preprocessing

PathMNIST, part of the MedMNIST v2 collection. Derived from a histopathology study of colorectal cancer: 28x28 RGB image patches extracted from hematoxylin and eosin (H&E) stained histology slides, downsampled from an original resolution of 224x224. Source: NCT-CRC-HE-100K (train/validation split) and CRC-VAL-HE-7K (test split, from a different clinical center; not loaded by this project). Only the train split is used.

Samples: 89,996 Features: 2,352 Sparsity: 0.0%

Missing values: No

Preprocessing: scale_unit_range

profile.json shows dense data (0.01% zeros) with values in [0, 255], and DATA_DESCRIPTION.md says these are raw 8-bit RGB pixel intensities from 28x28x3 H&E patches. That is a fixed, natural bound shared by every feature, so rescaling to [0, 1] puts all 2,352 pixels on the same footing without changing their relative contrast. Judgment call: standardize was the alternative, but it would give low-variance pixels (e.g. uniformly bright background regions) the same weight as informative ones, which the shared 0-255 scale already avoids. No count normalization or log1p: these are intensities, not counts. No select_top_variance: n_features (2,352) is far below n_samples (89,996), so dimensionality is not a problem that justifies discarding pixels.

Methods and hyperparameters

pca (role: general_purpose, status: reused)

Hyperparameters: `n_components=50`

Required linear baseline. `n_components = min(50, floor(89996/2), 2352) = 50`.

Neighboring pixels and the three color channels of the same pixel are strongly correlated in natural images, so a few dozen components should capture most variance; the sidecar reports exactly how much and how many a 90% target would need. PCA is cheap at 89,996 samples.

sparse_pca (role: general_purpose, status: reused)

Hyperparameters: `n_components=20, random_state=0`

Required PCA variant. `n_samples = 89,996` is far above the ~5,000 threshold, so Kernel PCA (a 89,996 x 89,996 kernel matrix) is not feasible without subsampling; Sparse PCA runs on the full data. Its sparse loadings also suit image data, since each component uses a limited set of pixels/channels, which is easier to interpret than dense PCA loadings. Judgment call: `n_components = 20` rather than PCA's 50, because method-selection/SKILL.md measured about 23 minutes for 50 components on this exact data shape and advises keeping it modest; 20 still spans the leading structure.

Alternative was subsampling to ~5,000 and running Kernel PCA, rejected because it would analyse only about 6% of the images.

umap (role: visualization_only, status: reused)

Hyperparameters: `n_components=2, n_neighbors=30, min_dist=0.1, random_state=0`

Visualization only. `n_samples = 89,996` is above 20,000, so `n_neighbors` is raised to 30 to keep a broader local neighborhood in a dataset this large; `min_dist = 0.1` is UMAP's default. The 9 tissue labels are used only to color it and for the label-silhouette sanity check. Never used for any computation.

Methods and hyperparameters (cont.)

tsne (role: visualization_only, status: reused)

Hyperparameters: n_components=2, perplexity=30, random_state=0

Second visualization layout, included for a specific reason: CLAUDE.md requires a second setting before trusting a structural claim from a manifold method, and only one UMAP artifact per method name is supported, so t-SNE (a different objective on the same preprocessed data) is the cross-check. A class-separation pattern is only described in the report if both UMAP and t-SNE show it. Perplexity 30 (sklearn default) is tiny relative to 89,996 samples; judgment call, since larger perplexity would emphasise more global structure but raises run time substantially at this size. Expected to be slow at ~90k points; still run on the full data rather than a subsample to keep one shared input across methods. Visualization only.

Quantitative metrics

method	trustworthiness	variance explained (PCs used)	PCs for 90% variance	label silhouette
pca	0.9820	0.7907	178	-0.0662
sparse_pca	0.9435	0.7154	>20	-0.0533
umap	0.8127			-0.0604
tsne	0.8347			-0.0336

How to read the metrics

Trustworthiness (`sklearn.manifold.trustworthiness`, 10 neighbours)

For each sample, takes its 10 nearest neighbours in the embedding and penalizes any that were not close neighbours in the original preprocessed data, weighting each penalty by how far down the original neighbour ranking that point sat. Ranges 0 to 1; 1 means the embedding introduces no false neighbours. It measures local neighbourhood fidelity only, not preservation of global distances. Computed against the data after the plan's preprocessing steps; on a random 5,000-sample subset when the dataset is larger than that (see Limitations).

Variance explained (PCs used)

The fraction of the total variance in the preprocessed data captured by the components actually kept in the embedding. For PCA, the sum of those components' explained-variance ratios (each component's variance divided by the total). For Sparse PCA, whose components are not orthogonal, no such per-component decomposition exists; instead this is the variance captured by the subspace the components span (the components orthogonalized via QR, then variance summed along that orthonormal basis) – a valid but different construction, marked in the sidecar JSON. Ranges 0 to 1. Blank for methods neither applies to.

PCs for 90% variance

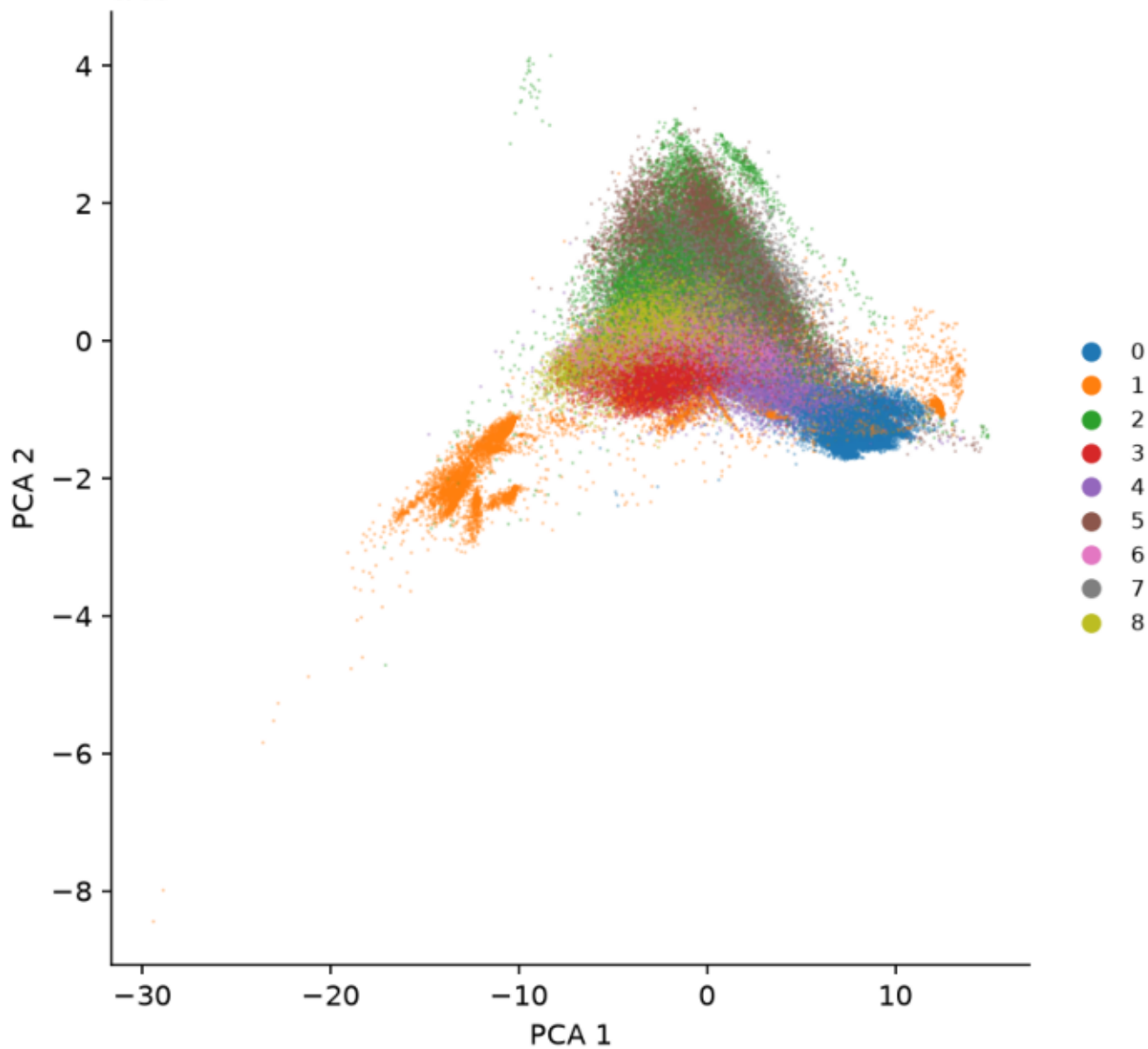
The smallest number of components whose cumulative explained variance reaches 90%. For PCA, read off a diagnostic fit of up to $\min(4 \times n_{\text{components}}, 200)$ components, deeper than what the embedding uses, so this can exceed the PCs actually kept. For Sparse PCA there is no deeper fit (it is too expensive to refit just for diagnostics); this is bounded by the components actually requested. Either way, a leading '>' means 90% was not reached within what was fit, so the true number is larger than shown. The scree plots show the full curve.

Label silhouette

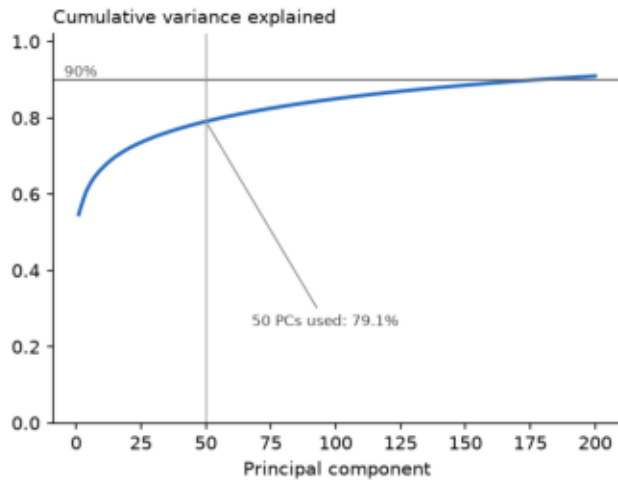
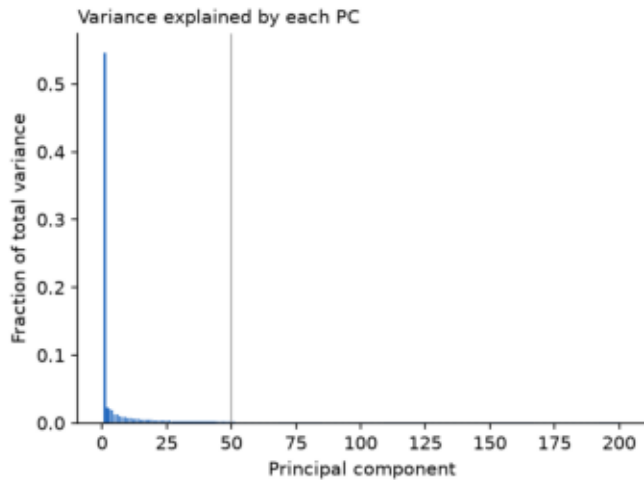
Mean silhouette coefficient (`sklearn.metrics.silhouette_score`) of the dataset's ground-truth class labels in the embedding space, using all embedding dimensions. Ranges -1 to 1: near 1 means points sit close to their own class and far from others, near 0 means classes overlap, negative means points are on average nearer another class than their own. A supervised sanity check only; labels were never used to build an embedding or to tune any hyperparameter.

pathmnist: pca

n=89,996, colored by ground-truth labels, showing dims 1-2
of 50

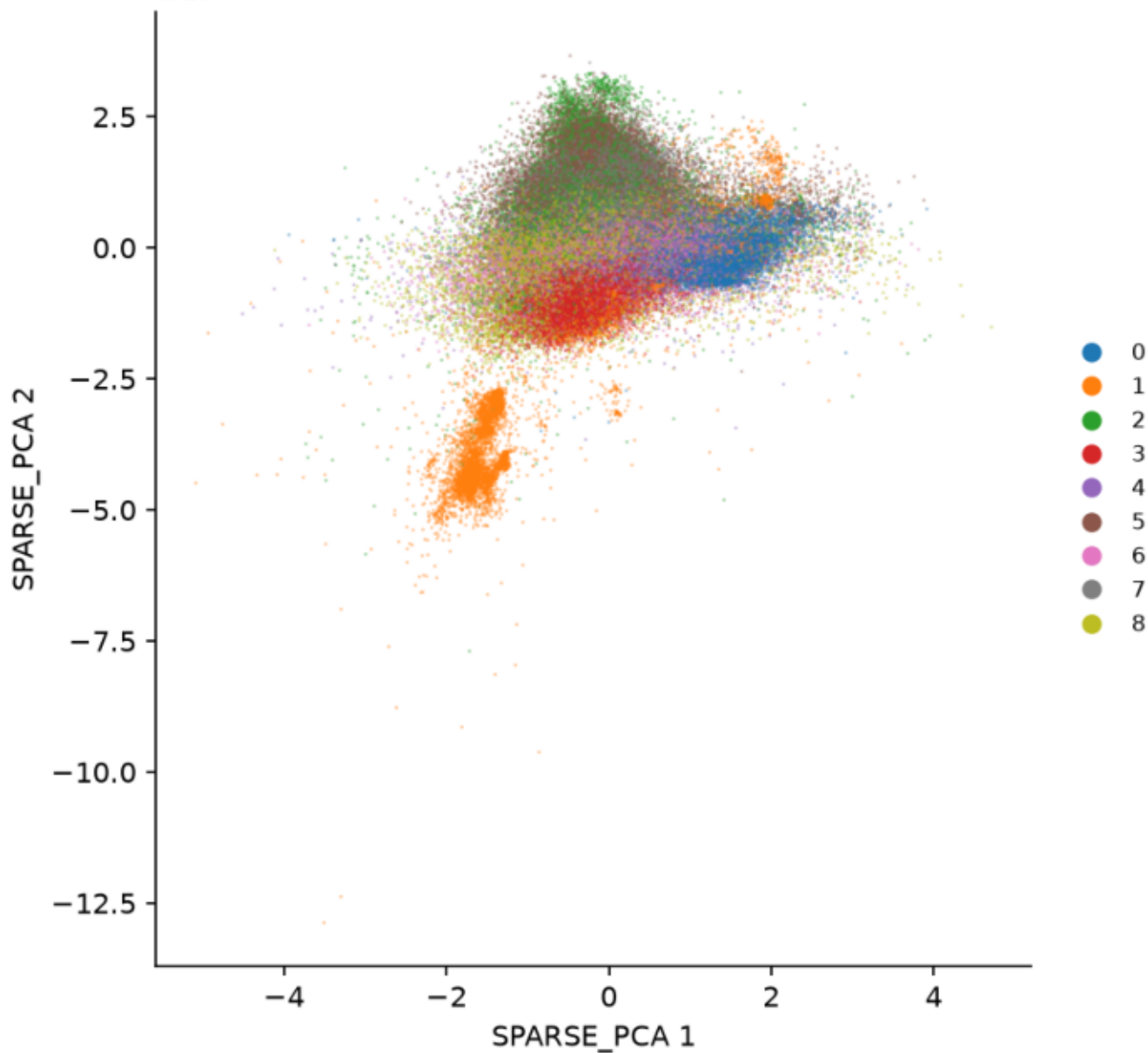


pathmnist: pca scree plot

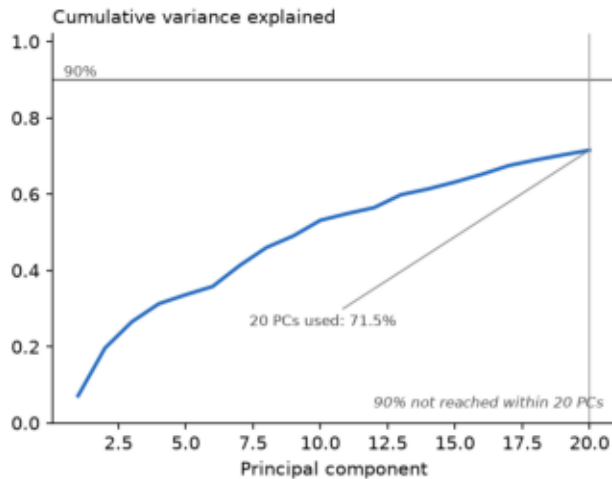
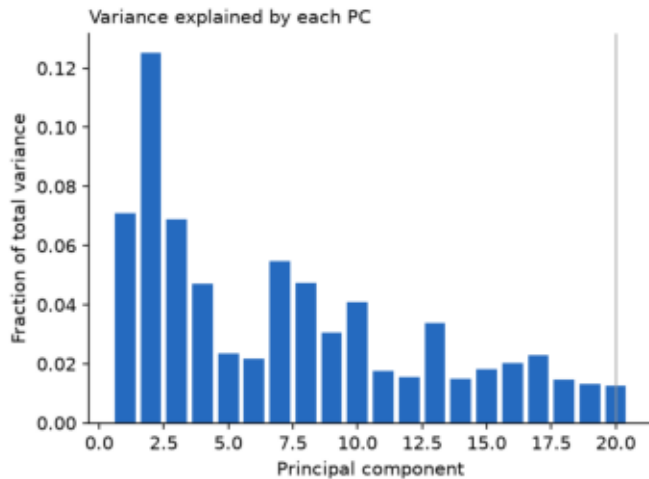


pathmnist: sparse_pca

n=89,996, colored by ground-truth labels, showing dims 1-2
of 20

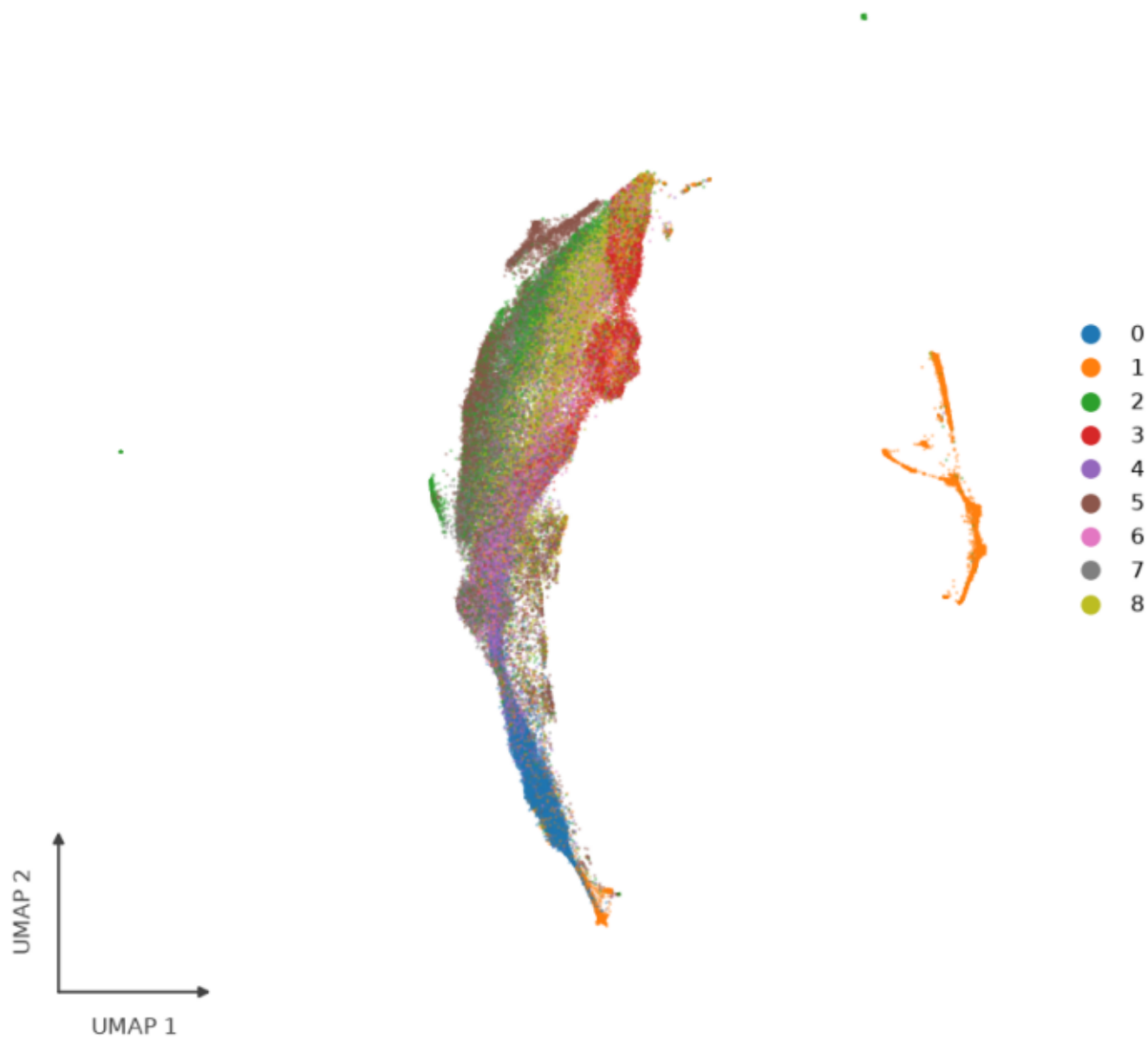


pathmnist: sparse_pca scree plot



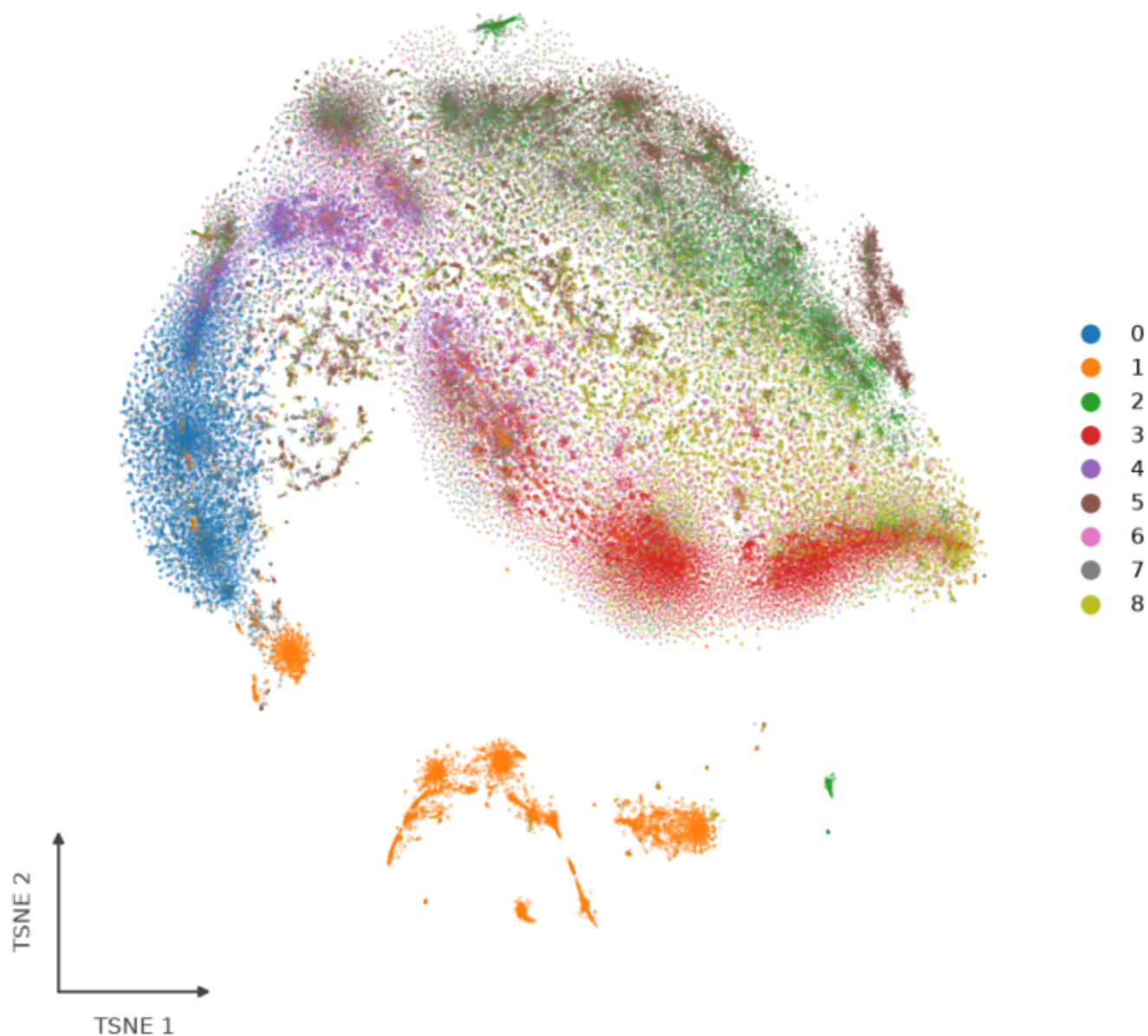
pathmnist: umap

n=89,996, colored by ground-truth labels



pathmnist: tsne

n=89,996, colored by ground-truth labels



Findings and interpretation

What each method produced

All four methods in the plan ran successfully on all 89,996 images after one shared preprocessing step (rescaling pixel intensities from [0, 255] to [0, 1]). No method needed a retry or a fallback. The seed is 0 throughout. Because the dataset is large, trustworthiness and the label silhouette were both computed on a random subset of 5,000 images, not on the full set.

PCA (50 components) captured 0.7907 of the total variance. The first component alone accounts for 0.5465; the next ones are much smaller (0.0229, 0.0209, 0.0185), and the scree plot shows one dominant component followed by a long, slowly decaying tail. Reaching 90% of the variance takes 178 components. So one direction carries over half of the pixel variance, while the rest is spread across many small directions. What that first direction represents was not tested here (the pipeline does not inspect loadings); in image data a single dominant component often tracks overall brightness, but that is a possibility, not a finding.

Sparse PCA was run with 20 components (reduced from 50 for run time, as recorded in the plan). Its components aren't orthogonal, so unlike PCA there's no per-component eigendecomposition; the variance-explained numbers instead measure the variance captured by the subspace the 20 components span (see the report's metric definitions), which still gives a scree plot and a comparable headline number, just not a strictly decreasing per-component curve. By that measure the 20 components captured 0.7154 of the total variance, comparable to PCA's 0.7907 from more than twice as many components; 90% was not reached within the 20 components fit (reported as >20, a true lower bound, not an estimate of how many more it would take, since a deeper diagnostic fit isn't run for Sparse PCA given the cost already measured at this size).

Trustworthiness (10 nearest neighbors) was 0.9820 for PCA (50 dimensions), 0.9435 for Sparse PCA (20 dimensions), 0.8347 for t-SNE (2 dimensions) and 0.8127 for UMAP (2 dimensions). These are not a ranking: the embeddings have different numbers of dimensions, and more dimensions leave more room to preserve neighborhoods. The fair comparison is between the two 2D methods, where t-SNE scores slightly higher on this measure.

Findings and interpretation (cont.)

Label sanity check

The 9 tissue labels were used only to color plots and to compute a silhouette score in each embedding. The label silhouette was slightly negative for every method: -0.0662 for PCA, -0.0533 for Sparse PCA, -0.0336 for t-SNE and -0.0604 for UMAP. A value near zero or below means that, measured by distances in these embeddings, images are on average about as close to images of other labels as to images of their own label. This says the unsupervised embeddings of raw pixels do not arrange the 9 labels into compact, well-separated groups overall. It does not mean the tissue types are indistinguishable: the figures show that some labels do separate while most overlap, and a single average score hides that difference.

What the figures show

The labels are stored as integers 0 to 8. DATA_DESCRIPTION.md lists the 9 tissue names but does not state which integer maps to which name, so this report refers to labels by number only rather than guessing the mapping.

Label 1 is the clearest separate group. It sits apart from the main body of points in PCA and Sparse PCA (at the low end of the second component and, in PCA, also the first), forms its own detached structure in UMAP, and forms several separate islands in t-SNE. Since this shows up in both linear methods and in both visualization methods, it is the most robust structural observation in this analysis.

Label 0 occupies its own region at one end of the main body in all four figures: at the high end of the first component in PCA and Sparse PCA, at the lower tip of the elongated UMAP shape, and as a distinct arm on the left of the t-SNE layout. It touches the rest of the data rather than being detached.

Label 3 is concentrated in a recognisable region in every figure (the lower-middle part of the PCA and Sparse PCA clouds, one side of the UMAP shape, and a dense band near the bottom of the main t-SNE mass), but it borders and partly overlaps neighbouring labels.

The remaining labels (2, 4, 5, 6, 7 and 8) overlap heavily in all four embeddings. Some show local concentrations in t-SNE (for example small label-5 and label-2 patches), but no method separates them cleanly. A few very small, tight groups of label-2 points sit away from the main mass in UMAP and t-SNE; their size and meaning cannot be read from these plots.

Findings and interpretation (cont.)

How the methods relate

PCA and Sparse PCA are the general-purpose embeddings, the ones whose coordinates could reasonably be used for further quantitative work; their 2D views look broadly alike, with the same labels separating. UMAP and t-SNE are pictures only. They were run together so that no structural statement rests on a single layout, and they agree on the points above: label 1 detached, label 0 at one end, label 3 in a concentrated region, and heavy overlap among the rest. Neither was used for any computation beyond its own trustworthiness and label-silhouette scores.

What cannot be concluded

These embeddings are built from raw 28x28 pixel intensities. DATA_DESCRIPTION.md notes that the downsampling from 224x224 loses fine histological texture and that pixel similarity does not imply tissue similarity, so heavy label overlap here is a statement about this pixel representation, not about how separable the tissue types are in principle. The 2D distances, island sizes and shapes in UMAP and t-SNE carry no quantitative meaning. Both scores were computed on a 5,000-image subset. Sparse PCA used fewer components than PCA (20 against 50), so their trustworthiness scores are not directly comparable either. Each method was run with one seed; the UMAP/t-SNE agreement is the only cross-check on the visual structure. Kernel PCA, MDS, Isomap and Diffusion Maps were not run because their $N \times N$ cost is impractical at 89,996 images.

Limitations

Trustworthiness was computed on a random subset of 5,000 of 89,996 samples (it is $O(N^2)$ and infeasible at full size) for: `pca`, `sparse_pca`, `umap`, `tsne`. Treat those scores as estimates, not exact full-dataset values.

The label silhouette column is a supervised sanity check only, computed on the embedding's own dimensions; it was never used to tune any hyperparameter.

The label silhouette was scored on a fixed-seed random subset of 5,000 of 89,996 samples (it is $O(N^2)$) for: `pca`, `sparse_pca`, `umap`, `tsne`. Treat it as an estimate, not an exact full-dataset value.

`sparse_pca`: 'variance explained' and 'PCs for 90%' come from the variance captured by the subspace the (non-orthogonal) components span, not a per-component eigendecomposition like PCA's; see the metric definitions above. Also bounded by the components actually requested, not a deeper diagnostic fit, since refitting deeper is too expensive for this method at this dataset's size.

No adversarial critique/revision pass was performed for this run (critic subagent not yet built).